

SECTION 16.1

4. Compute the vector assigned to the points $P = (1, 1, 0)$ and $Q = (2, 1, 2)$ by the vector fields $\mathbf{e}_r$, $\frac{\mathbf{e}_r}{r}$, and $\frac{\mathbf{e}_r}{r^2}$.

$$\vec{e}_r = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{(x, y, z)}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\vec{e}_r}{r} = \frac{\mathbf{r}}{|\mathbf{r}|^2}$$

$$\frac{\vec{e}_r}{r^2} = \frac{\mathbf{r}}{|\mathbf{r}|^3}$$

• for point 'P' $(1, 1, 0)$

$$\vec{F}_1 = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{(1, 1, 0)}{\sqrt{2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$\vec{F}_2 = \frac{\mathbf{r}}{|\mathbf{r}|^2} = \frac{(1, 1, 0)}{(\sqrt{2})^2} = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$\vec{F}_3 = \frac{\mathbf{r}}{|\mathbf{r}|^3} = \frac{(1, 1, 0)}{(\sqrt{2})^3} = \left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, 0\right)$$

for point Q: $(2, 1, 2)$

$$\vec{F}_1 = \frac{(2, 1, 2)}{3} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

$$\vec{F}_2 = \frac{(2, 1, 2)}{3^2} = \left(\frac{2}{9}, \frac{1}{9}, \frac{2}{9}\right)$$

$$\vec{F}_3 = \frac{(2, 1, 2)}{3^3} = \left(\frac{2}{27}, \frac{1}{27}, \frac{2}{27}\right)$$

21. A river 200 meters wide is modeled by the region in the xy -plane given by $-100 \leq x \leq 100$. The velocity vector field on the surface

of the river is given by $\mathbf{F} = \left\langle \frac{-x}{20}, 20 - \frac{x^2}{1000} \right\rangle$ in meters per second (m/s). Determine the coordinates of those points that have the maximum speed.

$$|\mathbf{F}| = \sqrt{\frac{x^2}{10^4} - \frac{3x^2}{80} + 400}$$

$$\frac{d|\mathbf{F}|}{dx} = \frac{1}{25000} x^3 - \frac{3}{40} x$$

$$x=0 \quad x = \pm 25\sqrt{30}$$

So,

$$\frac{3x^2}{2500} = \frac{-3}{40}$$

at $x=0$, it is negative while it is positive at $x = \pm 25\sqrt{30}$

Hence velocity is max at $x=0$
velocity here is $\langle 0, 20 \rangle$

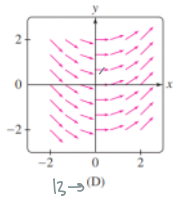
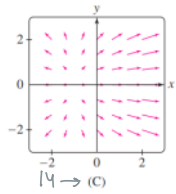
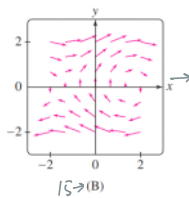
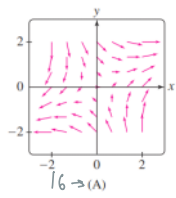
In Exercises 13–16, match each of the following planar vector fields with the corresponding plot in Figure 12.

13. $\mathbf{F} = (2, x)$

14. $\mathbf{F} = (2x + 2, y)$

15. $\mathbf{F} = (y, \cos x)$

16. $\mathbf{F} = (x + y, x - y)$



In Exercises 31–37, prove the identities assuming that the appropriate partial derivatives exist and are continuous.

34. $\text{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \text{curl}(\mathbf{F}) - \mathbf{F} \cdot \text{curl}(\mathbf{G})$ $\mathbf{F} = \langle M, N, P \rangle$ $\mathbf{G} = \langle A, B, C \rangle$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix} = \mathbf{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ N & P \end{vmatrix} - \mathbf{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ M & P \end{vmatrix} + \mathbf{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ M & N \end{vmatrix} = \mathbf{i} \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) - \mathbf{j} \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right) + \mathbf{k} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$\text{curl } \mathbf{G} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A & B & C \end{vmatrix} = \mathbf{i} \left(\frac{\partial C}{\partial y} - \frac{\partial B}{\partial z} \right) - \mathbf{j} \left(\frac{\partial C}{\partial x} - \frac{\partial A}{\partial z} \right) + \mathbf{k} \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y} \right)$$

L.H.S.:-

$$\mathbf{F} \times \mathbf{G} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ M & N & P \\ A & B & C \end{vmatrix} = \mathbf{i} (NC - PB) - \mathbf{j} (MC - AP) + \mathbf{k} (MB - NA)$$

- let,

$$H_1 = NC - PB$$

$$H_2 = MC - AP$$

$$H_3 = MB - NA$$

so,

$$\text{div}(\mathbf{H}) = \frac{\partial H_1}{\partial x} + \frac{\partial H_2}{\partial y} + \frac{\partial H_3}{\partial z}$$

$$\text{div}(\mathbf{H}) = \frac{\partial (NC - PB)}{\partial x} + \frac{\partial (MC - AP)}{\partial y} + \frac{\partial (MB - NA)}{\partial z}$$

applying product rule:- $\frac{\partial uv}{\partial x} = u \cdot \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x}$

$$\frac{\partial (NC - PB)}{\partial x} = \frac{\partial}{\partial x} NC - \frac{\partial}{\partial x} PB = \left(N \frac{\partial C}{\partial x} + C \frac{\partial N}{\partial x} \right) - \left(P \frac{\partial B}{\partial x} + B \frac{\partial P}{\partial x} \right)$$

$$\frac{\partial (MC - AP)}{\partial y} = \frac{\partial}{\partial y} MC - \frac{\partial}{\partial y} AP = \left(C \frac{\partial M}{\partial y} + M \frac{\partial C}{\partial y} \right) - \left(A \frac{\partial P}{\partial y} + P \frac{\partial A}{\partial y} \right)$$

$$\frac{\partial}{\partial z} (MB - NA) = \frac{\partial}{\partial z} MB - \frac{\partial}{\partial z} NA = \left(M \frac{\partial B}{\partial z} + B \frac{\partial M}{\partial z} \right) - \left(N \frac{\partial A}{\partial z} + A \frac{\partial N}{\partial z} \right)$$

$$\text{div}(\mathbf{F} \times \mathbf{G}) = \left(N \frac{\partial C}{\partial x} + C \frac{\partial N}{\partial x} \right) - \left(P \frac{\partial B}{\partial x} + B \frac{\partial P}{\partial x} \right) + \left(C \frac{\partial M}{\partial y} + M \frac{\partial C}{\partial y} \right) - \left(A \frac{\partial P}{\partial y} + P \frac{\partial A}{\partial y} \right) + \left(M \frac{\partial B}{\partial z} + B \frac{\partial M}{\partial z} \right) - \left(N \frac{\partial A}{\partial z} + A \frac{\partial N}{\partial z} \right)$$

for right hand side:-

$$= \mathbf{G} \cdot \text{curl}(\mathbf{F}) - \mathbf{F} \cdot \text{curl}(\mathbf{G})$$

$$= (A, B, C) \cdot \left(\left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right), \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right), \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \right) - (M, N, P) \cdot \left(\left(\frac{\partial C}{\partial y} - \frac{\partial B}{\partial z} \right), \left(\frac{\partial C}{\partial x} - \frac{\partial A}{\partial z} \right), \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y} \right) \right)$$

$$= \left[A \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) - B \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right) + C \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \right] - \left[M \left(\frac{\partial C}{\partial y} - \frac{\partial B}{\partial z} \right) - N \left(\frac{\partial C}{\partial x} - \frac{\partial A}{\partial z} \right) + P \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y} \right) \right]$$

$$= \left(A \cdot \frac{\partial P}{\partial y} - A \frac{\partial N}{\partial z} - B \frac{\partial P}{\partial x} + B \frac{\partial M}{\partial z} + C \frac{\partial N}{\partial x} - C \frac{\partial M}{\partial y} - M \frac{\partial C}{\partial y} + M \frac{\partial B}{\partial z} + N \frac{\partial C}{\partial x} - N \frac{\partial A}{\partial z} - P \frac{\partial B}{\partial x} + P \frac{\partial A}{\partial y} \right)$$

As LHS = RHS, Hence proved!

35. If f is a scalar function, then $\text{div}(f\mathbf{F}) = f \text{div}(\mathbf{F}) + \mathbf{F} \cdot \nabla f$.

$$\text{Let } \mathbf{F} = (F_1, F_2, F_3)$$

$$\text{div}(f(\mathbf{F})):-$$

$$f \cdot \mathbf{F} = (fF_1, fF_2, fF_3)$$

$$\text{div}(f \cdot \mathbf{F}) = \left(\frac{\partial}{\partial x} fF_1, \frac{\partial}{\partial y} fF_2, \frac{\partial}{\partial z} fF_3 \right)$$

applying product rule-

$$= \left(F_1 \frac{\partial f}{\partial x} + f \frac{\partial F_1}{\partial x} \right) + \left(f \frac{\partial F_2}{\partial y} + F_2 \frac{\partial f}{\partial y} \right) + \left(f \frac{\partial F_3}{\partial z} + F_3 \frac{\partial f}{\partial z} \right)$$

$$= F_1 \frac{\partial f}{\partial x} + f \frac{\partial F_1}{\partial x} + f \frac{\partial F_2}{\partial y} + F_2 \frac{\partial f}{\partial y} + f \frac{\partial F_3}{\partial z} + F_3 \frac{\partial f}{\partial z}$$

re-arranging:-

$$= \left(F_1 \frac{\partial f}{\partial x} + F_2 \frac{\partial f}{\partial y} + F_3 \frac{\partial f}{\partial z} \right) + \left(f \frac{\partial F_1}{\partial x} + f \frac{\partial F_2}{\partial y} + f \frac{\partial F_3}{\partial z} \right)$$



$\mathbf{F} \cdot \nabla f$



$f \cdot \text{div}(\mathbf{F})$

So,

$$= \mathbf{F} \cdot \nabla f + f \cdot \text{div}(\mathbf{F}) \quad \text{Hence proved!}$$

In Exercises 39–45, find a potential function for the vector field $\mathbf{F}$ by inspection or show that one does not exist.

42. $\mathbf{F} = \langle 2xyz, x^2z, x^2yz \rangle$

Checking if Potential function exists or not :-

let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$

$M = 2xyz$

$N = x^2z$

$P = x^2yz$

$\frac{\partial M}{\partial y} = 2xz$

$\frac{\partial M}{\partial x} = 2yz$

$\frac{\partial N}{\partial x} = 2xz$

$\frac{\partial P}{\partial x} = 2xyz$

$\frac{\partial M}{\partial z} = 2xy$

$\frac{\partial N}{\partial z} = x^2$

$\frac{\partial P}{\partial y} = x^2z$

Condition:-

$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \frac{\partial P}{\partial x} = \frac{\partial M}{\partial z}, \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$x^2z = x^2z, 2xyz = 2xy, 2xz = 2xz$

↑

condition false

Potential function do not exist.

44. $\mathbf{F} = \langle 2xe^{x^2}, 0, e^{x^2} \rangle$

$M = 2xe^{x^2}$

$N = 0$

$P = e^{x^2}$

$\frac{\partial M}{\partial y} = 0$

$\frac{\partial M}{\partial z} = 2e^{x^2} \cdot x$

$$\left| \begin{array}{l} \frac{\partial N}{\partial x} = 0 \\ \frac{\partial N}{\partial z} = 0 \end{array} \right| \quad \left| \begin{array}{l} \frac{\partial P}{\partial x} = 2xe^{x^2} \\ \frac{\partial P}{\partial y} = 0 \end{array} \right|$$

as all conditions are true-

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial z}, \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$

Now,

$\frac{\partial f}{\partial x} = M$

$\frac{\partial f}{\partial x} = 2xe^{x^2}$

$$\frac{\partial f}{\partial x} = 2xz e^{x^2}$$

integrate wrt x:-

$$f = 2 \int x z e^{x^2}$$

$$f = z \int e^{x^2} \cdot 2x$$

$$f = z e^{x^2} + g(y, z)$$

$$\frac{\partial f}{\partial y} = 0 + g'(y, z)$$

$$0 = g'(y, z)$$

integrating wrt y:-

$$0 = g(y, z) + h(z)$$

$$-h(z) = g(y, z)$$

Now

$$f = z e^{x^2} + (-h(z))$$

derivate wrt z

$$\frac{\partial f}{\partial z} = e^{x^2} - h'(z)$$

$$e^{x^2} = e^{x^2} - h'(z)$$

$$h'(z) = 0$$

integrating wrt z:-

$$h(z) = C$$

So,

$$f = z e^{x^2} + C$$

Ans

45. $\mathbf{F} = (yz \cos(xyz), xz \cos(xyz), xy \cos(xyz))$.

$$M = yz \cos(xyz) \quad N = xz \cos(xyz) \quad P = xy \cos(xyz)$$

$$\begin{aligned} \frac{\partial M}{\partial y} &= z(-\sin(nyz))xz \quad \checkmark & \frac{\partial N}{\partial z} &= x(-\sin(nyz)) \cdot ny \quad \checkmark \\ \frac{\partial M}{\partial z} &= y(-\sin(nyz))xy & \frac{\partial N}{\partial n} &= z(-\sin(nyz)) \cdot yz \quad \checkmark \\ \frac{\partial P}{\partial n} &= y(-\sin(nyz)) \cdot yz & \frac{\partial P}{\partial y} &= x(-\sin(nyz))xz \quad \checkmark \end{aligned}$$

Conditions: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial n}$, $\frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}$, $\frac{\partial M}{\partial z} = \frac{\partial P}{\partial n}$

$\uparrow$ $\uparrow$ $\uparrow$
 false false false

So, No potential function exist.

SECTION 16.2

3. Let $\mathbf{F} = \langle y^2, x^2 \rangle$, and let C be the curve $y = x^{-1}$ for $1 \leq x \leq 2$, oriented from left to right.

(a) Calculate $\mathbf{F}(\mathbf{r}(t))$ and $d\mathbf{r} = \mathbf{r}'(t) dt$ for the parametrization of C given by $\mathbf{r}(t) = (t, t^{-1})$.

(b) Calculate the dot product $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$ and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

$$\mathbf{r}(t) = (t, t^{-1})$$

$$a) \quad \mathbf{F}(\mathbf{r}(t)) = ((t^{-1})^2, t^2) = \left(\frac{1}{t^2}, t^2\right)$$

$$\mathbf{r}'(t) = (1, -t^{-2})$$

$$b) \quad \int_C \mathbf{F} \cdot d\mathbf{r} = \int \left(\frac{1}{t^2}, t^2\right) \cdot (1, -t^{-2}) \cdot dt$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int \left(\frac{1}{t^2} - \frac{t^2}{t^2}\right) \cdot dt$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^2 \left(\frac{1}{t^2} - 1\right) \cdot dt = \int_1^2 \frac{1}{t^2} \cdot dt - \int_1^2 dt$$

$$= \int_1^2 t^{-2} \cdot dt - \int_1^2 dt = -t^{-1} \Big|_1^2 - t \Big|_1^2$$

$$= [(-2^{-1}) - (-1^{-1})] - (2 - 1)$$

$$= -0.5 + 1 - (1)$$

$$= 0.5 - 1 = \boxed{-0.5} \text{ units}$$

$$= -0.5 + 1 - (1) \\ = 0.5 - 1 = \boxed{-0.5 \text{ dm}}$$

In Exercises 9–16, compute $\int_C f \, ds$ for the curve specified.

9. $f(x, y) = \sqrt{1 + 9xy}$, $y = x^3$ for $0 \leq x \leq 1$

let $x=t$ $0 \leq t \leq 1$

$$r(t) = \langle t, t^3 \rangle$$

$$r'(t) = \langle 1, 3t^2 \rangle$$

$$|r'(t)| = \sqrt{1^2 + (3t^2)^2} = \sqrt{1 + 9t^4}$$

$$f(r(t)) = \sqrt{1 + 9(t)(t)^3} = \sqrt{1 + 9t^4}$$

Now,

$$\begin{aligned} \int_C f \, ds &= \int_0^1 f(r(t)) \cdot |r'(t)| \, dt \\ &= \int_0^1 (\sqrt{1 + 9t^4}) \, dt \end{aligned}$$

$$= \int_0^1 1 \, dt + 9 \int_0^1 t^4 \, dt$$

$$= t \Big|_0^1 + 9 \frac{t^5}{5} \Big|_0^1$$

$$= (1 - 0) + 9 \left(\frac{1}{5} - \frac{0}{5} \right)$$

$$= 1 + 9/5 = \boxed{14/5 \text{ dm}}$$

10. $f(x, y) = \frac{y^3}{x^7}$, $y = \frac{1}{4}x^4$ for $1 \leq x \leq 2$

let $x=t$

$$r(t) = \left\langle t, t^4/4 \right\rangle$$

$$r(t) = \left(t, t^{4/4} \right)$$

$$f(r(t)) = \left(\frac{t^4}{4} \right)^3 \times \frac{1}{t^7} = \frac{t^{12}}{64} \times \frac{1}{t^7} = \frac{t^5}{64}$$

$$r'(t) = \left(1, 4 \frac{t^3}{4} \right) = (1, t^3)$$

$$\|r'(t)\| = \sqrt{1^2 + t^{3 \times 2}} = \sqrt{1 + t^6}$$

$$= \int_C f \cdot dr = \int_1^2 f(r(t)) \cdot \|r'(t)\| \cdot dt$$

$$= \int_1^2 \frac{t^5}{64} \cdot \sqrt{1+t^6} \cdot dt$$

$$= \frac{1}{64} \int_1^2 t^5 \sqrt{1+t^6} \cdot dt$$

$$= \frac{1}{64} \int_1^{64} \sqrt{1+u} \cdot du$$

$$= \frac{1}{64} \int_1^{64} (1+u)^{1/2} \cdot du$$

$$= \frac{1}{64} \left[\frac{(1+u)^{3/2}}{3/2} \right]_1^{64}$$

$$= \frac{1}{32} \left[\frac{2}{3} (1+u)^{3/2} \right]_1^{64}$$

$$= \frac{1}{96} \left[(1+64)^{3/2} - (1+1)^{3/2} \right]$$

$$= \frac{1}{96} \left[\sqrt{274,625} - \sqrt{8} \right] = \frac{1}{96} \left[524.04 - 2.82 \right]$$

$$= \boxed{5.428} \text{ Am.}$$

12. $f(x, y, z) = 3x - 2y + z$, $r(t) = (2+t, 2-t, 2t)$
for $-2 \leq t \leq 1$

$$\begin{aligned} f(r(t)) &= 3(2+t) - 2(2-t) + 2t \\ &= 6+3t-4+2t+2t = \boxed{2+7t} \end{aligned}$$

Now,

$$r'(t) = (0+1, 0-1, 2) = (1, -1, 2)$$

Now, $r'(t) = (0+1, 0-1, 2) = (1, -1, 2)$

$$|r'(t)| = \sqrt{1^2 + 1^2 + 2^2} = \sqrt{6}$$

$$\begin{aligned} \int_C F \cdot dr &= \int_{-2}^1 (2+7t) \cdot \sqrt{6} \cdot dt \\ &= \int_{-2}^1 (2\sqrt{6} + 7\sqrt{6}t) \cdot dt \end{aligned}$$

$$= 2\sqrt{6} \int_{-2}^1 dt + 7\sqrt{6} \int_{-2}^1 t \cdot dt$$

$$= 2\sqrt{6} t \Big|_{-2}^1 + 7\sqrt{6} t^2 \Big|_{-2}^1$$

$$= 2\sqrt{6} (1 - (-2)) + 7\sqrt{6} (1^2 - (-2)^2)$$

$$= 2\sqrt{6} (3) + 7\sqrt{6} (-3)$$

$$= 6\sqrt{6} - 21\sqrt{6} = \boxed{15\sqrt{6}} \text{ Ans}$$

In Exercises 27–32, evaluate the line integral.

28. $\int_C ydx + zdy + xdz$, $r(t) = (2+t^{-1}, t^3, t^2)$ for $0 \leq t \leq 1$

$$x(t) = 2+t^{-1}$$

$$y(t) = t^3$$

$$z(t) = t^2$$

$$\frac{dx}{dt} = -t^{-2}$$

$$dx = -t^{-2} \cdot dt$$

$$\frac{dy}{dt} = 3t^2$$

$$dy = 3t^2 \cdot dt$$

$$\frac{dz}{dt} = 2t$$

$$dz = 2t \cdot dt$$

$$= \int_0^1 [t^3 \cdot (-t^{-2}) \cdot dt + t^2 (3t^2 \cdot dt) + (2+t^{-1}) \cdot 2t \cdot dt]$$

$$= \int_0^1 (-t^1) \cdot dt + \int_0^1 3t^4 \cdot dt + \int_0^1 (4t+2) \cdot dt$$

$$= -\frac{t^2}{2} \Big|_0^1 + 3 \frac{t^5}{5} \Big|_0^1 + 4 \frac{t^2}{2} \Big|_0^1 + 2 t \Big|_0^1$$

$$= -\frac{1}{2} (1^2 - 0^2) + \frac{3}{5} (1^5 - 0^5) + 2 (1^2 - 0^2) + 2 (1 - 0)$$

$$= -\frac{1}{2} + \frac{3}{5} + 2 + 2$$

$$= 1.5 + 2 + \frac{3}{5}$$

$$= \frac{7}{2} + \frac{3}{5}$$

$$= \frac{35 + 6}{10} = \frac{41}{10} \text{ Ans.}$$

32. $\int_C y^2 dx + z^2 dy + (1 - x^2) dz$, quarter of the circle of radius 1

$$\begin{aligned} x &= r \cos t \\ y &= r \sin t \\ z &= 0 \end{aligned} \quad 0 \leq t \leq \pi/2$$

$$\frac{dx}{dt} = -\sin(t) \quad dx = -\sin(t) \cdot dt$$

$$\frac{dy}{dt} = \cos(t) \quad dy = \cos(t) \cdot dt$$

$$\frac{dz}{dt} = 0$$

$$dz = 0 \cdot dt$$

$$F = \int_0^{\pi/2} \sin^2 t \cdot (-\sin(t)) \cdot dt$$

$$F = - \int_0^{\pi/2} \sin^3 t \cdot dt$$

$$F = - \int_0^{\pi/2} \sin t (1 - \cos^2 t) \cdot dt$$

$$F = - \int_0^{\pi/2} (\sin t - \sin t \cos^2 t) \cdot dt$$

$$F = - \int_0^{\pi/2} \sin t \cdot dt + \int_0^{\pi/2} \cos^2 t \cdot \sin t \cdot dt$$

$$F = + \cos(t) \Big|_0^{\pi/2} - \frac{\cos^3(t)}{3} \Big|_0^{\pi/2}$$

$$F = + (\cos(\pi/2) - \cos(0)) - \left[\frac{\cos^3(\pi/2)}{3} - \frac{\cos^3(0)}{3} \right]$$

$$F = -1 - \left[0 + \frac{1}{3} \right]$$

$$\boxed{F = -\frac{2}{3}} \text{ Ans.}$$

In Exercises 35 and 36, calculate the line integral of $\mathbf{F}(x, y, z) = \langle e^z, e^{x-y}, e^y \rangle$ over the given path.

36. The closed path ABCA in Figure 15

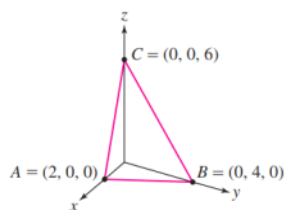


FIGURE 15

• for line AB:- $0 \leq t \leq 1$

$$\mathbf{r}(t) = \mathbf{A} + t(\overrightarrow{AB})$$

$$\mathbf{r}(t) = (2, 0, 0) + (-2, 4, 0)t$$

$$\mathbf{r}(t) = (2, 0, 0) + (-2t, 4t, 0)$$

$$\mathbf{r}(t) = (2+2t, 4t, 0)$$

• for line BC:- $0 \leq t \leq 1$

$$\begin{aligned} \mathbf{r}(t) &= \mathbf{B} + t\overrightarrow{BC} \\ &= (0, 4, 0) + (0, -4, 6)t \\ &= (0, 4-4t, 6t) \end{aligned}$$

• for line CA:- $0 \leq t \leq 1$

$$\mathbf{r}(t) = (0, 0, 6) + t(2, 0, -6)$$

$$\mathbf{r}(t) = (0, 0, 6) + (2t, 0, -6t)$$

$$\mathbf{r}(t) = (2t, 0, 6-6t)$$

• for vector line integral of line AB:

$$\mathbf{F} = \langle e^z, e^{x-y}, e^y \rangle$$

$$\mathbf{r}(t) = \langle 2+2t, 4t, 0 \rangle$$

$$r = \langle x, y, z \rangle$$

$$r(t) = \langle 2+2t, 4t, 0 \rangle$$

$$F(r(t)) = \langle 1, e^{2-2t}, e^{4t} \rangle$$

Now,

$$\int_c F \cdot T \, ds = \int_c \langle 1, e^{2-2t}, e^{4t} \rangle \cdot \langle 2+2t, 4t, 0 \rangle \cdot dt$$

$$= \int_0^1 (2+2t + 4t \cdot e^{2-2t} + 0) \cdot dt$$

$$= \int_0^1 2 \cdot dt + 2 \int_0^1 t \cdot dt + 4 \int_0^1 \frac{e^2}{e^{2t}} \cdot t \cdot dt$$

$$= 2t \Big|_0^1 + t^2 \Big|_0^1 + 4e^2 \int_0^1 e^{-2t} \cdot t \cdot dt$$

integrating by parts:-

$$\int u \cdot dv = uv - \int v \cdot du$$

$$\text{let } u = t \\ v = e^{(-2t)} \cdot dt$$

$$= t \left(-\frac{1}{2} e^{(-2t)} \right) - \int \left(-\frac{1}{2} e^{(-2t)} \right) \cdot dt$$

$$= -\frac{t}{2} e^{(-2t)} + \frac{1}{2} \int e^{-2t} \cdot dt$$

$$= \left(-\frac{t}{2} e^{-2t} - \frac{1}{4} e^{-2t} \right) \Big|_0^1$$

$$= 2(1-0) + (1^2-0^2) + 4e^2 \left[\left(-\frac{1}{2} e^{-2} - \frac{1}{4} e^{-2} \right) - \left(-\frac{1}{4} e^0 \right) \right]$$

$$= 2 + 1 + 4e^2 \left(-\frac{3}{4e^2} + \frac{1}{4} \right)$$

$$= 3 - 3 + \frac{e^2}{4} = \boxed{\frac{e^2}{4} \text{ Ans}}$$

• for vector integral of line $\overline{BC}$:

$$F = \langle e^z, e^{xy}, e^y \rangle$$

$$r(t) = \langle 0, 4-4t, 6t \rangle$$

$$F(r(t)) = \langle e^{4t}, e^{4t-4}, e^{4-4t} \rangle$$

$$\int_C F \cdot T \, ds = \int_0^1 \langle e^{4t}, e^{4t-4}, e^{4-4t} \rangle \cdot \langle 0, 4-4t, 6t \rangle \, dt$$

$$= \int_0^1 (0 + (4-4t) \cdot e^{4t-4} + 6t(e^{4-4t})) \, dt$$

$$= \int_0^1 [4e^{4t-4} - 4e^{4t-4} \cdot t + 6e^{4-4t} \cdot t] \, dt$$

$$= 4 \int_0^1 e^{4t-4} \, dt - 4 \int_0^1 e^{4t-4} \cdot t \, dt + 6 \int_0^1 e^{4-4t} \cdot t \, dt$$

$$= \frac{4}{e^4} \int_0^1 e^{4t} \, dt - \frac{4}{e^4} \int_0^1 e^{4t} \cdot t \, dt + \frac{6}{e^4} \int_0^1 e^{4t} \cdot t \, dt$$

$$= \frac{4}{e^4} \int_0^1 e^{4t} \, dt + \frac{2}{e^4} \int_0^1 e^{4t} \cdot t \, dt$$

$$u = t \quad dv = e^{4t}$$

$$= \frac{4}{e^4} \cdot \left. \frac{e^{4t}}{4} \right|_0^1 + \frac{2}{e^4} \left(t \cdot \frac{e^{4t}}{4} - \int \frac{e^{4t}}{4} \, dt \right) \Big|_0^1$$

$$= \frac{4}{e^4} \cdot \left(\frac{e^4}{4} - \frac{e^0}{4} \right) + \frac{2}{e^4} \left(\frac{e^4 \cdot 1}{4} - \frac{e^{4t}}{16} \right) \Big|_0^1$$

$$= \left[\frac{e^4 - 1}{e^4} \right] + \frac{2}{e^4} \left[\left(\frac{e^4}{4} - \frac{e^4}{16} \right) - \left(0 - \frac{1}{16} \right) \right]$$

$$= 1 - \frac{1}{e^4} + \frac{2}{e^4} \left[\left(\frac{4e^4 - e^4}{16} \right) + \frac{1}{16} \right]$$

$$= 1 - \frac{1}{e^4} + \frac{2}{e^4} \left(\frac{3e^4}{16} + \frac{1}{16} \right)$$

$$= 1 - \frac{1}{e^4} + \frac{6}{16} + \frac{1}{8e^4}$$

$$= \frac{16+6}{16} + \frac{1}{8e^4} = \boxed{\frac{11}{8} + \frac{1}{8e^4}} \text{ km}$$

• for vector integral of line $\vec{CA}$:-

$$F = \langle e^z, e^{x-y}, e^y \rangle$$

$$r(t) = \langle 2t, 0, 6-6t \rangle$$

$$(e^z, e^{x-y}, e^y)$$

$$F(r(t)) = \langle e^{6-6t}, e^{2t}, 1 \rangle$$

$$\begin{aligned}
\int_C \mathbf{F} \cdot d\mathbf{s} &= \int_0^1 \left\langle \frac{e^t}{e^{6t}}, e^{2t}, 1 \right\rangle \cdot \langle 2t, 0, 6-6t \rangle \cdot dt \\
&= \int_0^1 \left\langle 2t \cdot \frac{e^t}{e^{6t}} + 0 + 6-6t \right\rangle \cdot dt \\
&= \int_0^1 2t \cdot \frac{e^t}{e^{6t}} \cdot dt + 6 \int_0^1 1 \cdot dt - 6 \int_0^1 t \cdot dt \\
&= 2e^6 \int_0^1 e^{-6t} \cdot t \cdot dt + 6t \Big|_0^1 - 6 \frac{t^2}{2} \Big|_0^1 \\
&= 2e^6 \left[t \cdot \frac{e^{-6t}}{-6} + \frac{1}{6} \int e^{-6t} \cdot dt \right] + 6(1-0) - 6 \left(\frac{1^2}{2} - 0 \right) \\
&= 2e^6 \left[\left(-\frac{t}{6} e^{-6t} - \frac{1}{36} e^{-6t} \right) \right]_0^1 + 6 - 3 \\
&= 2e^6 \left[\left(-\frac{e^{-6}}{6} - \frac{1}{36} e^{-6} \right) - \left(0 - \frac{1}{36} e^{-6} \right) \right] + 3 \\
&= 6e^6 + 2e^6 \left[\frac{-6e^{-6} - e^{-6}}{36} + \frac{1}{36} \right] \\
&= 6e^6 - \frac{17e^6}{36} + \frac{e^6}{18} \\
&= \frac{108e^6 + e^6 - 17e^6}{36} = \boxed{6.05e^6 - 7/18} \text{ km}
\end{aligned}$$

Now add:-

$$\begin{aligned}
&= \left(6.05e^6 - 7/18 \right) + \left(\frac{11}{8} + \frac{9}{8e^4} \right) + \frac{e^2}{4} \\
&= \boxed{\frac{109}{18} e^6 - \frac{7}{18} + \frac{11}{8} + \frac{9}{8e^4} + \frac{e^2}{4} \text{ km.}}
\end{aligned}$$

38. Find the value of $\int_{C'} \mathbf{F} \cdot d\mathbf{r}$, where C' is the path that traverses the loop C_2 four times in the clockwise direction.

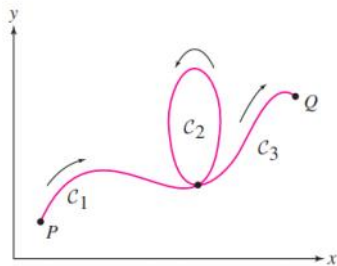


FIGURE 16

We know that $\int_C \mathbf{F} \cdot d\mathbf{r} = 5$, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = 8$, $\int_{C_3} \mathbf{F} \cdot d\mathbf{r} = 8$

So,
 $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = ?$

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + \int_{C_3} \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

$$8 + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + 8 = 5$$

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = -11 \quad \text{for 4 time clockwise}$$

$$-11 \times 4 = \boxed{44 \text{ Ans}}$$

44. Calculate the total mass of a metal tube in the helical shape $\mathbf{r}(t) = (\cos t, \sin t, t^2)$ (distance in centimeters) for $0 \leq t \leq 2\pi$ if the mass density is $\delta(x, y, z) = \sqrt{z}$ g/cm.

$$|\mathbf{r}(t)| = \sqrt{t^2} = t$$

$$\mathbf{r}'(t) = (-\sin t, \cos t, 2t)$$

$$|\mathbf{r}'(t)| = \sqrt{(\sin^2 t) + \cos^2 t + 4t^2} = \sqrt{1 + 4t^2} = \sqrt{1 + 4t^2}$$

• finding scalar line integral-

$$\int_0^{2\pi} t \cdot \sqrt{1 + 4t^2} \cdot dt$$

let $u = 1 + 4t^2$
 $du = 8t$

$$\text{let } u = 1 + 4t^2$$

$$\frac{du}{dt} = 8t$$

$$\frac{du}{8} = t \cdot dt$$

$$= \int_0^{2\pi} \sqrt{u} \cdot \frac{du}{8}$$

$$= \frac{1}{8} \frac{u^{3/2}}{3/2} \Big|_0^{2\pi} = \frac{2}{24} u^{3/2} \Big|_0^{2\pi}$$

• putting value of u :-

$$= \frac{1}{12} \left[u^{3/2} \right]_0^{2\pi}$$

$$= \frac{1}{12} \left[(1 + 4t^2)^{3/2} \right]_0^{2\pi}$$

$$= \frac{1}{12} \left(\left[1 + 4(2\pi)^2 \right]^{3/2} - \left[1 \right]^{3/2} \right) = 166.85 \text{ grams.}$$

46. Find the total charge on the curve $\mathbf{r}(t) = (\sin t, \cos t, \sin^2 t)$ in centimeters for $0 \leq t \leq \frac{\pi}{8}$ assuming a charge density of $\delta(x, y, z) = xy(y^2 - z)$ (in units of 10^{-6} C/cm).

$$F(\mathbf{r}(t)) = (\sin t \cdot \cos t) (\cos^2 t - \sin^2 t)$$

$$= \sin t \cdot \cos t \cdot \cos 2t \quad \therefore \cos^2 t - \sin^2 t = \cos 2t$$

$$\mathbf{r}'(t) = (\cos t, -\sin t, \sin 2t)$$

$$|\mathbf{r}'(t)| = \sqrt{\cos^2 t + \sin^2 t + (\sin 2t)^2} = \sqrt{1 + \sin^2 2t}$$

$$\int_C F \cdot \mathbf{T} \, ds = \int_C (\cos^3 t \cdot \sin t \cdot \cos 2t) \cdot \sqrt{1 + \sin^2 t} \, dt$$

$$= \int_0^{\pi/8} \cos t \cdot \sin t \cdot \cos 2t \cdot \sqrt{1 + \sin^2 t} \, dt$$

$$= \frac{1}{2} \int_0^{\pi/8} \sin 2t \cdot \cos 2t \cdot \sqrt{1 + \sin^2 t} \, dt$$

$$\begin{aligned} \text{let } 2t = y \\ \frac{dy}{dt} = 2 \\ \frac{dy}{2} = dt \\ &= \frac{1}{2} \int_0^{\pi/4} \sin y \cdot \cos y \cdot \sqrt{1 + \sin^2\left(\frac{y}{2}\right)} \cdot \frac{dy}{2} \\ &= \frac{1}{4} \int_0^{\pi/4} \sin y \cdot \cos y \sqrt{1 + \sin^2\left(\frac{y}{2}\right)} \cdot dy \end{aligned}$$

by calculator:-

$$\begin{aligned} &= \frac{1}{4} \times (0.258) = 0.064 \times 10^{-6} \text{ C/cm} \\ &= 6.4 \times 10^{-8} \text{ C/cm} \quad \text{Ans.} \end{aligned}$$

55. Figure 21 shows a force field $\mathbf{F}$.

(a) Over which of the two paths, ADC or ABC , does $\mathbf{F}$ perform less work?

(b) If you have to work against $\mathbf{F}$ to move an object from C to A , which of the paths, CBA or CDA , requires less work?

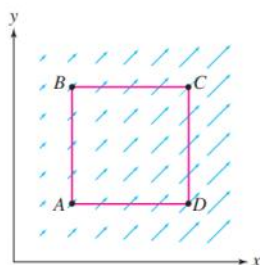


FIGURE 21

a) As we move along path ADC we see the general force direction aligns with the path. Same is the case for path ABC , but, force appears to be stronger along the path ADC because of bigger arrows, so along the path ADC , more work is done. So, PATH ABC performs less work.

b) PATH CBA will require less work because less force is exerted against it.

56. Verify that the work performed along the segment $\overrightarrow{PQ}$ by the constant vector field $\mathbf{F} = \langle 2, -1, 4 \rangle$ is equal to $\mathbf{F} \cdot \overrightarrow{PQ}$ in these cases:

(a) $P = (0, 0, 0)$, $Q = (4, 3, 5)$

(b) $P = (3, 2, 3)$, $Q = (4, 8, 12)$

a) as we know that $\text{work} = \mathbf{F} \cdot \mathbf{d}$ so,

$$d = Q - P = (4, 3, 5) - (0, 0, 0) = (4, 3, 5)$$

Now,

$$\begin{aligned} \text{work} &= F \cdot d = F \cdot \vec{PQ} = \langle 2, -1, 4 \rangle \cdot (4, 3, 5) \\ &= 8 + (-3) + (20) = \boxed{25 \text{ units}} \end{aligned}$$

b) Same for his part $PQ = Q - P = (4, 8, 12) - (3, 2, 3) = (1, 6, 9)$

$$\text{work} = F \cdot d = \langle 2, -1, 4 \rangle \cdot (1, 6, 9) = 2 - 6 + 36 = \boxed{32 \text{ units}}$$

58. Note that a curve C in polar form $r = f(\theta)$ is parametrized by $\mathbf{r}(\theta) = (f(\theta) \cos \theta, f(\theta) \sin \theta)$ because the x - and y -coordinates are given by $x = r \cos \theta$ and $y = r \sin \theta$.

(a) Show that $\|\mathbf{r}'(\theta)\| = \sqrt{f(\theta)^2 + f'(\theta)^2}$.

(b) Evaluate $\int_C (x - y)^2 ds$, where C is the semicircle in Figure 22 with polar equation $r = 2 \cos \theta$, $0 \leq \theta \leq \frac{\pi}{2}$.

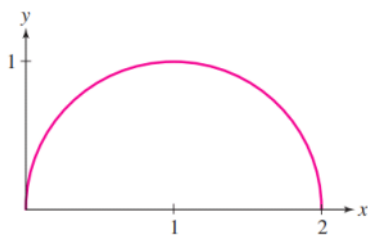


FIGURE 22 Semicircle $r = 2 \cos \theta$.

a) $\mathbf{r}'(\theta) = (\cos \theta f'(\theta) - f(\theta) \sin \theta, f'(\theta) \sin \theta + f(\theta) \cos \theta)$

$$\|\mathbf{r}'(\theta)\| = \sqrt{(\cos \theta f'(\theta) - f(\theta) \sin \theta)^2 + (f'(\theta) \sin \theta + f(\theta) \cos \theta)^2}$$

$$= \sqrt{[(\cos \theta f'(\theta))^2 - 2f'(\theta)f(\theta)\cos \theta \sin \theta + (f(\theta) \sin \theta)^2] + [(f'(\theta) \sin \theta)^2 + 2f(\theta)f'(\theta)\sin \theta \cos \theta + (f(\theta) \cos \theta)^2]}$$

$$= \sqrt{(\cos \theta (f'(\theta))^2 + (f(\theta) \sin \theta)^2 + (f'(\theta) \sin \theta)^2 + (f(\theta) \cos \theta)^2)}$$

$$= \sqrt{f'(\theta)^2 (\sin^2 \theta + \cos^2 \theta) + f(\theta)^2 (\sin^2 \theta + \cos^2 \theta)} = \sqrt{f'(\theta)^2 + f(\theta)^2}$$

Here proved.

b) $\int_C (x - y)^2 ds$, $r = 2 \cos \theta$, $0 \leq \theta \leq \frac{\pi}{2}$.

$$r = 2 \cos \theta$$

$$r^2 = 2r \cos \theta$$

So,

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$(x-1)^2 + y^2 = 0 \quad \text{— equation}$$

$$\text{radius} = 1$$

$$x = \cos \theta$$

$$f(\theta) = 2 \cos \theta$$

$$y = \sin \theta$$

$$f'(\theta) = -2 \sin \theta$$

from part A, we now find

$$ds = \sqrt{(2 \cos \theta)^2 + (2 \sin \theta)^2} = 2 \sqrt{1} = 2 \cdot d\theta$$

Now,

$$= \int_0^{\pi/2} (\cos \theta - \sin \theta)^2 \cdot 2 \, d\theta$$

$$= 2 \int_0^{\pi/2} (\cos^2 \theta - 2 \cos \theta \sin \theta + \sin^2 \theta) \, d\theta$$

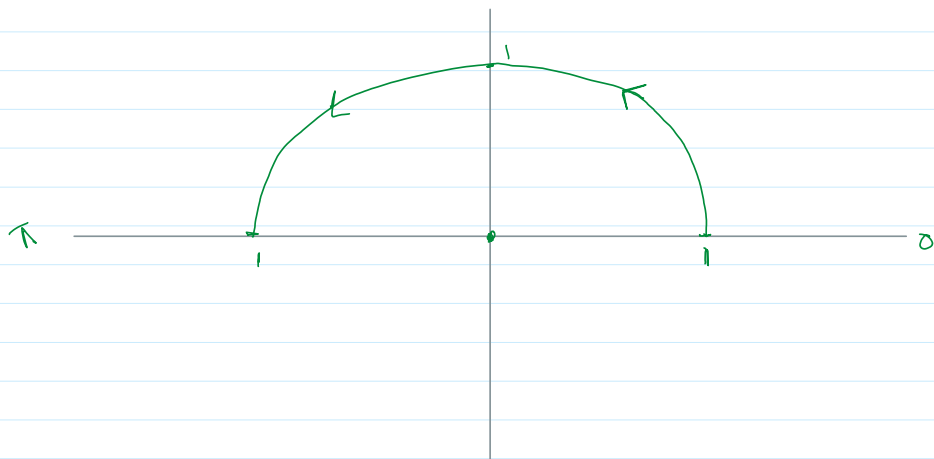
$$= 2 \left[\int_0^{\pi/2} \cos^2 \theta \cdot d\theta - 2 \int_0^{\pi/2} \cos \theta \sin \theta \cdot d\theta + \int_0^{\pi/2} \sin^2 \theta \cdot d\theta \right]$$

$$= 2 \left[\left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) \Big|_0^{\pi/2} - 2 \frac{\sin^2 \theta}{2} \Big|_0^{\pi/2} + \left(\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right) \Big|_0^{\pi/2} \right]$$

$$= \pi \quad \text{answer}$$

SECTION 16.3

4. $\mathbf{F}(x, y) = \langle \cos y, -x \sin y \rangle$, $f(x, y) = x \cos y$; upper half of the unit circle centered at the origin, oriented counterclockwise



$$\nabla f = F(x, y)$$

$$f(x, y) = x \cos y$$

$$f_x = \frac{\partial}{\partial x} x \cos y = \cos y$$

$$f_y = \frac{\partial}{\partial y} x \cos y = -x \sin y$$

$$\nabla f(x, y) = \langle \cos y, -x \sin y \rangle = F(x, y)$$

so, f is a potential function of F .

- Apply the integral theorem

$$\int_c F \cdot dr = f(r(b)) - f(r(a))$$

$$a = (-1, 0) \quad b = (1, 0)$$

so, $f(x, y) = x \cos y$

at $(-1, 0)$

$$f(-1, 0) = (-1) \cos(0) = -1$$

at $(1, 0)$

$$f(1, 0) = 1$$

$$\int_C F \cdot dr = f(1,0) - f(-1,0) = 1 - (-1) = 2$$

$$\boxed{\int_C F \cdot dr = -2} \text{ because anti clockwise direction}$$

18. Evaluate

$$\oint_C \sin x \, dx + z \cos y \, dy + \sin y \, dz$$

where C is the ellipse $4x^2 + 9y^2 = 36$, oriented clockwise.

$$4x^2 + 9y^2 = 36$$

Divide by 36 b.s.

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$a=3 \quad b=2$$

$$x = 3 \cos \theta$$

$$y = 2 \sin \theta$$

$$z = 0$$

$$\frac{dx}{d\theta} = -3 \sin \theta$$

$$dx = -3 \sin \theta \cdot d\theta$$

$$dy = 2 \cos \theta \cdot d\theta$$

$$dz = 0$$

$$\oint_C \langle \sin x \cdot dx, 0, \sin y \cdot dz \rangle$$

$$\oint_C P \, dx + Q \, dy = - \iint_D \left(x \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot dA$$

$$P = \sin x \quad Q = 0$$

$$\frac{\partial P}{\partial y} = 0$$

$$x \frac{\partial Q}{\partial x} = 0$$

$$= \iint_D (0 - 0) \cdot dA = \boxed{0}$$

Ans.

20. $f = zy + xy + xz$, $\mathbf{r}_1 = \langle t, t, t \rangle$ for $0 \leq t \leq 1$, $\mathbf{r}_2 = \langle t, t^2, t^3 \rangle$ for $0 \leq t \leq 1$

$$\mathbf{F} = \nabla f = \langle y+z, z+x, y+x \rangle$$

$$\mathbf{c}'(t) = \langle 1, 1, 1 \rangle$$

$$\mathbf{F}(\mathbf{c}_1(t)) = \langle 2t, 2t, 2t \rangle$$

$$\int_0^1 \langle 2t, 2t, 2t \rangle \cdot \langle 1, 1, 1 \rangle dt = \int_0^1 6t dt = 3t^2 \Big|_0^1 = 3$$

$$\mathbf{c}'_2(t) = \langle 1, 2t, 3t^2 \rangle$$

$$\mathbf{F}(\mathbf{c}_2(t)) = \langle t^2+t^3, t^3+t, t^2+t \rangle$$

$$\begin{aligned} \int_0^1 \langle t^2+t^3, t^3+t, t^2+t \rangle \cdot \langle 1, 2t, 3t^2 \rangle dt &= \int_0^1 (t^2+t^3+2t(t^3+t)+3t^4+3t^3) dt \\ &= \int_0^1 (3t^2+4t^3+5t^4) dt = \boxed{3} \end{aligned}$$

as both results matched so, integral along both paths is equal.

26. An electric dipole with dipole moment $p = 4 \times 10^{-5}$ C-m sets up an electric field (in newtons per coulomb)

$$\mathbf{F}(x, y, z) = \frac{kp}{r^5} \langle 3xz, 3yz, 2z^2 - x^2 - y^2 \rangle$$

where $r = (x^2 + y^2 + z^2)^{1/2}$ with distance in meters and $k = 8.99 \times 10^9$ N-m²/C². Calculate the work against $\mathbf{F}$ required to move a particle of charge $q = 0.01$ C from $(1, -5, 0)$ to $(3, 4, 4)$. Note: The force on q is $q\mathbf{F}$ newtons.

let potential energy = U

$$\text{work done} = U(A) - U(B)$$

$$U = \frac{kpqz}{r^3}$$

so,

at point A (1, -5, 0)

$$r_A = \sqrt{1^2 + (-5)^2 + 0^2} = \sqrt{26} \quad z_A = 0$$

$$U(A) = \frac{kpq(0)}{(\sqrt{26})^3} = 0$$

at B (3, 4, 4)

$$r_B = \sqrt{3^2 + 4^2 + 4^2} = \sqrt{41} \quad z_B = 4$$

$$U(B) = \frac{kpq(4)}{(\sqrt{41})^3} =$$

$$W = U_A - U_B = 0 - \frac{kpq(4)}{(41)^{3/2}} = \frac{4kpg}{(41)^{3/2}}$$

Now,

$$k = 8.99 \times 10^9 \quad p = 4 \times 10^{-5}, \quad q = 0.01$$

$$W = \frac{1.438 \times 10^4}{262.5} = \boxed{55 \text{ J}} \quad \text{Ans.}$$

Scalar electric potential of a dipole at (x, y, z)

$$V(x, y, z) = \frac{kpz}{r^3}$$

$$\text{so } U(x, y, z) = qV(x, y, z)$$

$$= \frac{kpqz}{r^3}$$

3. Relation between U , V , and q

- The dipole has an electric potential (voltage field):

$$V(x, y, z) = \frac{k \vec{p} \cdot \vec{r}}{r^3}$$
- For a dipole aligned along the z-axis, this reduces to:

$$V(x, y, z) = \frac{kpz}{r^3}$$
- The potential energy of a charge q in that potential is:

$$U(x, y, z) = qV(x, y, z)$$

That's what I called U .

$$p = (0, 0, p)$$

$$\vec{r} = (x, y, z)$$

$$\vec{p} \cdot \vec{r} = pz$$

28. An electron at rest at $P = (5, 3, 7)$ moves along a path ending at $Q = (1, 1, 1)$ under the influence of the electric field (in newtons per coulomb)

$$\mathbf{F}(x, y, z) = 400(x^2 + z^2)^{-1} \langle x, 0, z \rangle$$

(a) Find a potential function for $\mathbf{F}$.

(b) What is the electron's speed at point Q ? Use Conservation of Energy and the value $q_e/m_e = -1.76 \times 10^{11} \text{ C/kg}$, where q_e and m_e are the charge and mass on the electron, respectively.

a)

$$\frac{dv}{dx} = \frac{400x}{x^2 + z^2}$$

$$V(x, y, z) = \int \frac{400x}{x^2 + z^2} \cdot dx$$

$$200 \ln(x^2 + z^2) + g(y, z)$$

$$\frac{d}{dy} (200 \ln(x^2 + z^2)) = 0$$

$$\frac{d}{dz} (200 \ln(x^2 + z^2)) = 0$$

$$\frac{d}{dz} (200 \ln(x^2 + z^2)) = \frac{400z}{x^2 + z^2}$$

$$F_1 = 0$$

$$F_2 = \frac{400z}{x^2 + z^2}$$

$$\text{So, } g(y, z) = 0$$

$$V(x, y, z) = 200 \ln(x^2 + z^2)$$

$$b) V(1, 1, 1) - V(5, 3, 7) = 200 \ln(2) - 200 \ln(74) = -722.184$$

$$= \frac{mc^2 v^2}{2qe} = -722.184$$

$$V = \sqrt{722.184 \times 1.7 \times 10^{-18} \times 2} = 1.594 \times 10^7 \text{ m/s}$$

29. Let $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ be the vortex field. Determine

$\int_C \mathbf{F} \cdot d\mathbf{r}$ for each of the paths in Figure 20.

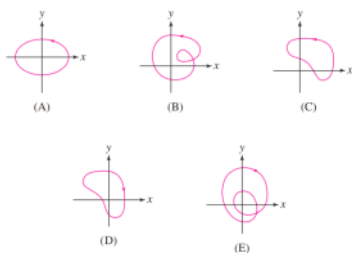


FIGURE 20

$$\leftarrow \int_C \mathbf{F} \cdot d\mathbf{r} = 2\pi n, \text{ where}$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

Path A) The path goes counter clockwise around the origin once
line integral = 2π

B) Goes counter clockwise around the origin once, line integral = 2π

C) The path doesn't go around origin so, line integral = 0

D) Goes clockwise around the origin once, line integral = -2π

e) cross curve clockwise and the origin twice, the integral = 4π

30. The vector field $\mathbf{F}(x, y) = \left\langle \frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right\rangle$ is defined on the domain $\mathcal{D} = \{(x, y) \neq (0, 0)\}$.

(a) Is $\mathcal{D}$ simply connected?

(b) Show that $\mathbf{F}$ satisfies the cross-partial condition. Does this guarantee that $\mathbf{F}$ is conservative?

(c) Show that $\mathbf{F}$ is conservative on $\mathcal{D}$ by finding a potential function.

(d) Do these results contradict Theorem 4?

$$\mathbf{F} = \left\langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \right\rangle \quad \mathcal{D} = \{(x, y) \neq (0, 0)\}$$

a) No, a hole exists at the origin with the open domain.

$$b) \frac{\partial f}{\partial y} = x(x^2+y^2)^{-1} = \frac{-2yx}{(x^2+y^2)^2}$$

$$\frac{\partial g}{\partial x} = y(x^2+y^2)^{-1} = \frac{-2yx}{(x^2+y^2)^2}$$

- The cross partial conditions are satisfied, so, No proof of $\mathbf{F}$ being conservative as Domain is not simply connected.

$$c) V(x, y) = \frac{1}{2} \ln(x^2+y^2)$$

$$\nabla V = \left\langle \frac{\partial V}{\partial x}, \frac{\partial V}{\partial y} \right\rangle = \left\langle \frac{x}{x^2+y^2}, \frac{y}{x^2+y^2} \right\rangle = \mathbf{F}$$

so, $\mathbf{F}$ is conservative.